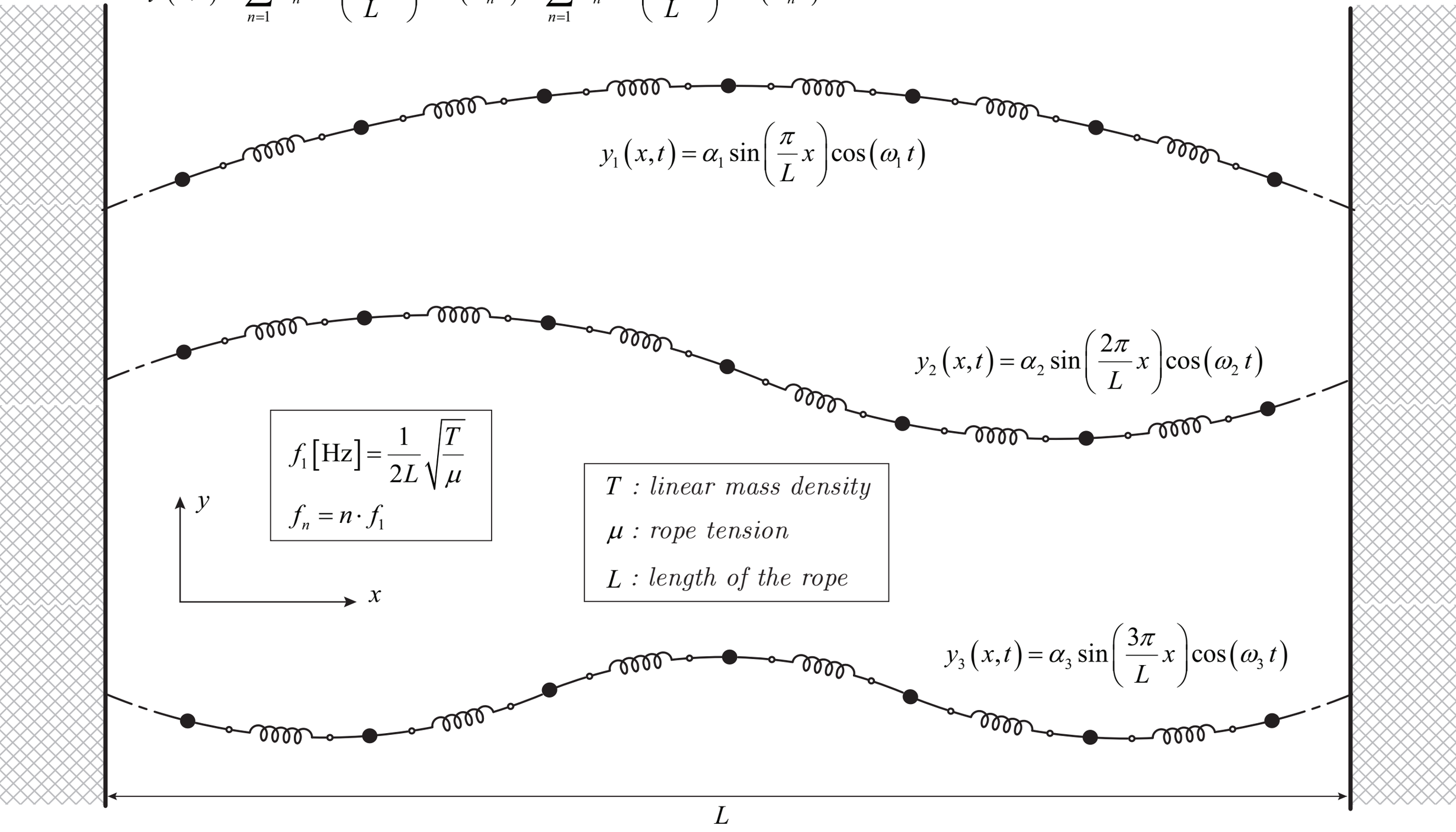$$y(x,t)=\sum_{n=1}^{\infty}\alpha_n \sin\left(\frac{n\pi}{L}x\right)\cos(\omega_n t)\approx\sum_{n=1}^{N}\alpha_n \sin\left(\frac{n\pi}{L}x\right)\cos(\omega_n t)$$